

EXAM 7 – SPRING 2014

1. (2 points)

Given the following information:

Cumulative Losses (\$000,000)

<u>Accident Year</u>	<u>Reported at 24 Months</u>	<u>Ultimate Loss</u>
2008	12	18
2009	10	16
2010	14	20
2011	12	18
2012	21	

An insurer writes annual policies that inception on January 1. Exposure and coverage levels were constant for 2008 through 2011. On January 1, 2012, policy coverage was expanded and pricing actuaries estimated the following:

- Loss amounts will increase by 25% due to the expanded coverage.
- 75% of ultimate losses are expected to be reported by 24 months, with a standard deviation of 8% of estimated ultimate loss.
- Standard deviation of accident year 2012 ultimate loss will be \$3 million.

a. (1.5 points)

Calculate the projected accident year 2012 ultimate loss using Bayesian credibility methods.

b. (0.5 point)

Explain why the least squares method is not appropriate for calculating the accident year 2012 ultimate loss.

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SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 1	
TOTAL POINT VALUE: 2	LEARNING OBJECTIVE: A1: Calculate unpaid claim estimates using credibility models.
SAMPLE ANSWERS	
Part a: 1.5 points	
<i>Sample Answer 1</i> X= loss reported at 24 months Y= Ultimate losses $L(x)=Z(x/d)+(1-Z)E[Y]$ $Z=\frac{VHM}{VHM+EVPV}$ $VHM=(E[D].\sigma(y))^2=((.75)(3))^2=5.0625$ $EVPV=Var(D)[Var(y)+E[y]^2]=(0.08)^2[3^2+[(1.25)(\frac{18+16+20+18}{4})]^2]=3.2976$ $Z=\frac{5.0625}{5.0625+3.2976}=.606$ $L(x)=(.606)(21/.75)+(1-.606)(22.5)=25.833$ million <i>Sample Answer 2</i> Y= ultimate losses $Z=\frac{VHM}{VHM+EVPV}$ $\sigma_y=3M$ $d=.75$ $\sigma_d=.08$ $VHM=d^2 \sigma_y^2$ $EVPV=\sigma_d^2[\sigma_y^2+E(y)^2]$ $E[y]=1.25 \times 18=22.5M$ $VHM=.75^2 3^2=5.0625$ $EVPV=.08^2[3^2+22.5^2]=3.2976$ $Z=\frac{5.0625}{5.0625+3.2976}=.6055$ Ultimate $=.6055(21/.75)+(1-.6055)(22.5)=25.83M$	
Part b: 0.5 point	
<i>Sample 1</i> The least squares method is appropriate when the distribution of loss is not changing year over year. Given the coverage expansion and change in 2012 loss distribution, we cannot use the least squares method. <i>Sample 2</i> Least squares method is appropriate when year to year fluctuation is mainly caused by randomness. However, in this case, there is a systematic shift in losses. As a result, we cannot use least squares.	

SAMPLE ANSWERS AND EXAMINER'S REPORT

EXAMINER'S REPORT
Part a
The candidate is expected to know a. Application of credibility b. Mechanics of the method To get full credit, the candidate has to get the final calculation right. Candidates did not score very well on this part. Only 36% received full credit. Part of the reason is that it involved a lot of calculation. Some candidates had difficulty calculating $E[y]$. Common errors include using EVPV (VHM) formula for VHM (EVPV), missing 25% increase in $E(y)$, using LDF to calculate x/d or $E(y)$, using standard deviation instead of variance in calculation. Some candidates used formula involving LDF for Z.
Part b
The candidate is expected to know strengths and weaknesses of the method. To get full credit, the candidate has to mention Least Squares method assumptions and why is it not the case in this question. Candidates performed well on this question; 90% of candidates received full credit. One common error was mentioning only part of the two components (least squares relies on stable patterns, major change in coverage distorts stable pattern).